

Running Time of BUILD MAX HEAP

Alg: BUILD-MAX-HEAP(A)

1. $n = \text{length}[A]$
 2. **for** $i \leftarrow \lfloor n/2 \rfloor$ **downto** 1
 3. **do** MAX-HEAPIFY(A, i, n)
- $O(\lg n)$ } $O(n)$

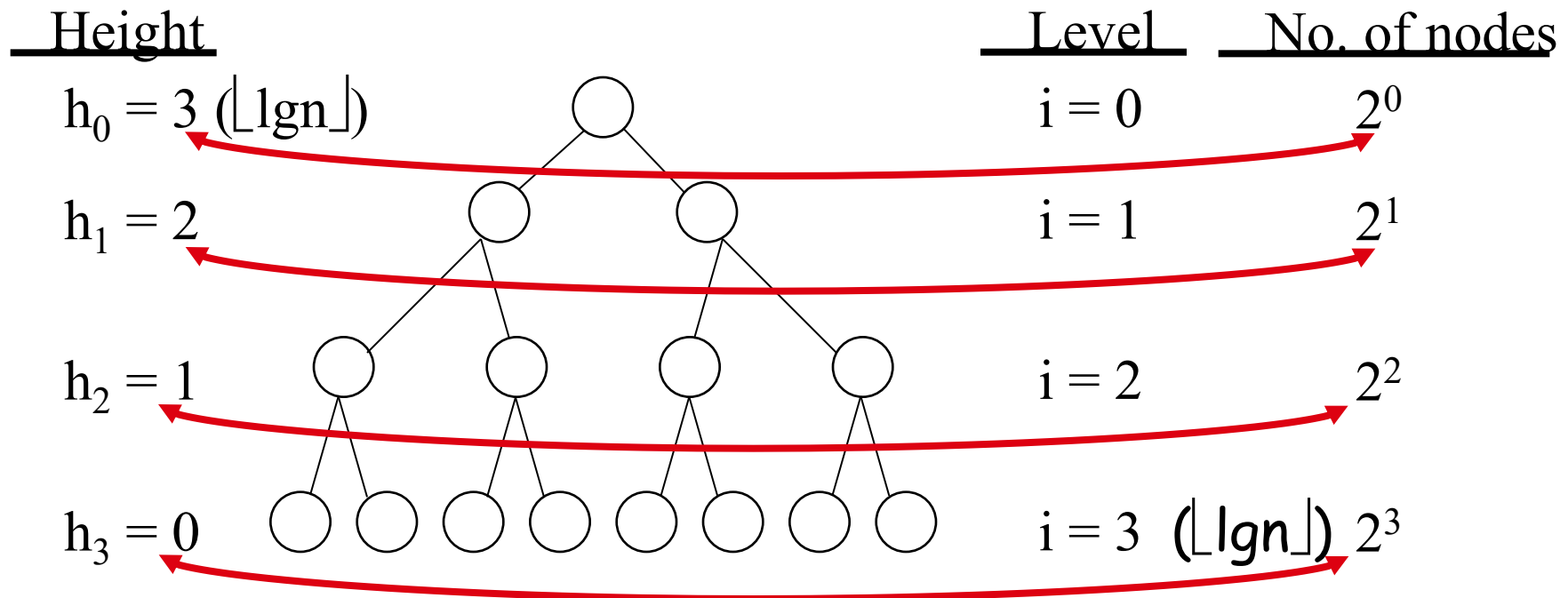
$\Rightarrow$ Running time: $O(n \lg n)$

- This is not an asymptotically tight upper bound

Running Time of BUILD MAX HEAP

- HEAPIFY takes $O(h) \Rightarrow$ the cost of HEAPIFY on a node i is proportional to the height of the node i in the tree

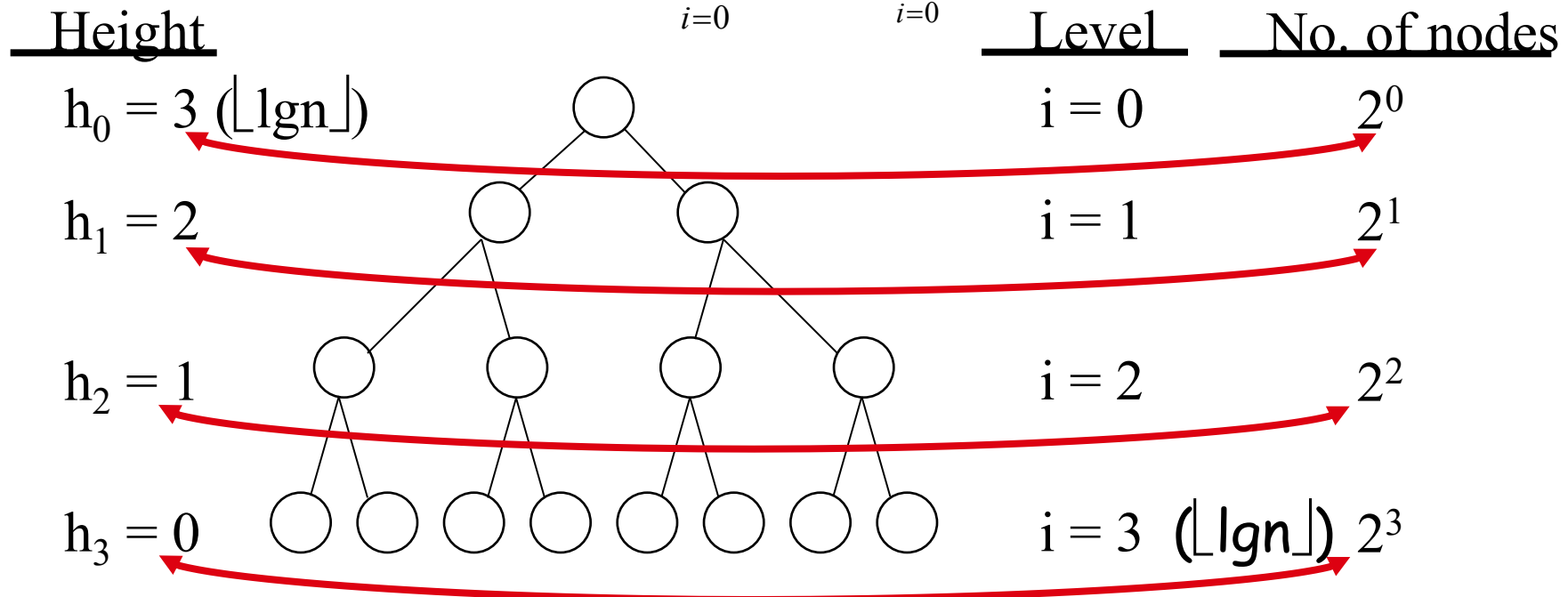
$$T(n) = O(n)$$



Running Time of BUILD MAX HEAP

- HEAPIFY takes $O(h) \Rightarrow$ the cost of HEAPIFY on a node i is proportional to the height of the node i in the tree

$$\Rightarrow T(n) = \sum_{i=0}^h n_i h_i = \sum_{i=0}^h 2^i (h - i) = O(n)$$



$h_i = h - i$ height of the heap rooted at level i

$n_i = 2^i$ number of nodes at level i

Running Time of BUILD MAX HEAP

$$T(n) = \sum_{i=0}^h n_i h_i \quad \text{Cost of HEAPIFY at level } i * \text{number of nodes at that level}$$

$$= \sum_{i=0}^h 2^i (h - i) \quad \text{Replace the values of } n_i \text{ and } h_i \text{ computed before}$$

$$= \sum_{i=0}^h \frac{h-i}{2^{h-i}} 2^h \quad \text{Multiply by } 2^h \text{ both at the nominator and denominator and write } 2^i \text{ as } \overline{2^{-i}}$$

$$= 2^h \sum_{k=0}^h \frac{k}{2^k} \quad \text{Change variables: } k = h - i$$

$$\leq n \sum_{k=0}^{\infty} \frac{k}{2^k} \quad \text{The sum above is smaller than the sum of all elements to } \infty \text{ and } h = \lg n$$

$$= O(n) \quad \text{The sum above is smaller than 2}$$

Running time of **BUILD-MAX-HEAP**: $T(n) = O(n)$

Running Time of BUILD MAX HEAP

- We can derive a tighter bound by observing that the time for MAX-HEAPIFY to run at a node varies with the height of the node in the tree, and the heights of most nodes are small.
- n-element heap has height $\lceil \lg n \rceil$
- at most $\lceil n/2^{h+1} \rceil$ nodes of any height h
- The time required by MAX-HEAPIFY when called on a node of height h is $O(h)$

Running Time of BUILD MAX HEAP

$$\sum_{h=0}^{\lfloor \lg n \rfloor} \left\lceil \frac{n}{2^{h+1}} \right\rceil O(h) = O \left(n \sum_{h=0}^{\lfloor \lg n \rfloor} \frac{h}{2^h} \right) .$$

We evaluate the last summation by substituting $x = 1/2$ in the formula (A.8) yielding

$$\begin{aligned} \sum_{h=0}^{\infty} \frac{h}{2^h} &= \frac{1/2}{(1 - 1/2)^2} \\ &= 2 . \end{aligned}$$

Thus, we can bound the running time of BUILD-MAX-HEAP as

$$\begin{aligned} O \left(n \sum_{h=0}^{\lfloor \lg n \rfloor} \frac{h}{2^h} \right) &= O \left(n \sum_{h=0}^{\infty} \frac{h}{2^h} \right) \\ &= O(n) . \end{aligned}$$

Hence, we can build a max-heap from an unordered array in linear time.